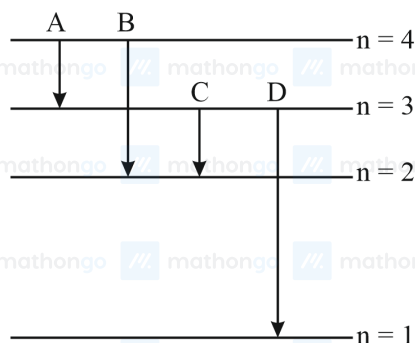


1. The energy levels of an hydrogen atom are shown below. The transition corresponding to emission of shortest wavelength is



[2023 (06 Apr Shift 1)]

- (1) D
 - (2) A
 - (3) B
 - (4) C
2. The radius of fifth orbit of Li^{++} is $\text{---} \times 10^{-12}$ m. Take: radius of hydrogen atom = $0.51 \text{ \AA}$
- [2023 (06 Apr Shift 1)]
3. The ratio of wavelength of spectral lines H_α and H_β in the Balmer series is $\frac{x}{20}$. The value of x is --- .
- [2023 (08 Apr Shift 2)]
4. The angular momentum for the electron in Bohr's orbit is L . If the electron is assumed to revolve in second orbit of hydrogen atom, then the change in angular momentum will be
- [2023 (10 Apr Shift 1)]
- (1) Zero
 - (2) $2L$
 - (3) L
 - (4) $\frac{L}{2}$
5. If $917 \text{ \AA}$ be the lowest wavelength of Lyman series then the lowest wavelength of Balmer series will be $\text{\AA}$.
- [2023 (10 Apr Shift 2)]
6. A monochromatic light is incident on a hydrogen sample in ground state. Hydrogen atoms absorb a fraction of light and subsequently emit radiation of six different wavelengths. The frequency of incident light is $x \times 10^{15}$ Hz. The value of x is ---
- (Given $h = 4.25 \times 10^{-15} \text{ eVs}$)
- [2023 (11 Apr Shift 1)]
7. The energy of He^+ ion in its first state is, (The ground state energy for the Hydrogen atom -13.6 eV):
- [2023 (11 Apr Shift 2)]
- (1) -27.2 eV
 - (2) -3.4 eV
 - (3) -13.6 eV
 - (4) -54.4 eV
8. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. The number of spectral lines emitted will be:
- [2023 (12 Apr Shift 1)]
- (1) 1
 - (2) 4
 - (3) 2
 - (4) 3
9. The radius of 2nd orbit of He^+ of Bohr's model is r_1 and that of fourth orbit of Be^{3+} is represented as r_2 . Now the ratio $\frac{r_2}{r_1}$ is $x : 1$. The value of x is --- .
- [2023 (13 Apr Shift 1)]
10. An atom absorbs a photon of wavelength 500 nm and emits another photon of wavelength 600 nm . The net energy absorbed by the atom in this process is $n \times 10^{-4} \text{ eV}$. The value of n is [Assume the atom to be stationary during the absorption and emission process] (Take $h = 6.6 \times 10^{-34} \text{ Js}$ and $c = 3 \times 10^8 \text{ m s}^{-1}$).
- [2023 (13 Apr Shift 2)]

11. As per given figure A , B and C are the first, second and third excited energy levels of hydrogen atom respectively. If the ratio of the two wavelengths (i.e. $\frac{\lambda_1}{\lambda_2}$) is $\frac{7}{4n}$, then the value of n will be

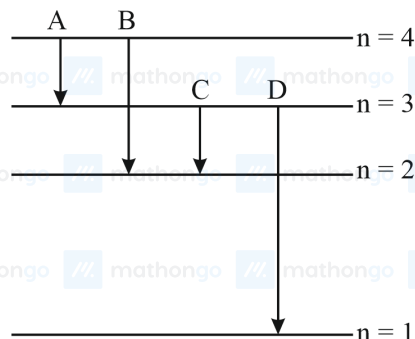


[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (1) 2. (425) 3. (27) 4. (3) 5. (3668) 6. (3) 7. (3) 8. (4)
9. (2) 10. (4125) 11. (5)

1. (1)



Wavelength of photons released during de-excitation is given by,

$$\frac{1}{\lambda} = R \left[\frac{1}{(n_1)^2} - \frac{1}{(n_2)^2} \right]$$

Clearly, shortest wavelength will correspond to highest energy difference in de-excitation, which in the given figure will correspond to transition from $n = 3$ to $n = 1$.

2. (425)

The formula to calculate the radius of Lithium-ion is given by

$$R_{Li} = R_H \times \frac{n^2}{Z} \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required radius of the ion.

$$R_{Li} = \frac{0.51 \times 25}{3} \times 10^{-10} \text{ m} \\ = 425 \times 10^{-12} \text{ m}$$

3. (27)

The formula to calculate the wavelength of H_α -line in Balmer series is given by

$$\frac{hc}{\lambda_\alpha} = 13.6 \left[\frac{1}{4} - \frac{1}{9} \right] \dots (1)$$

The formula to calculate the wavelength of H_β -line in Balmer series is given by

$$\frac{hc}{\lambda_\beta} = 13.6 \left[\frac{1}{4} - \frac{1}{16} \right] \dots (2)$$

Divide equation (2) by equation (1)

$$\Rightarrow \frac{\lambda_\alpha}{\lambda_\beta} = \frac{\frac{1}{4} - \frac{1}{16}}{\frac{1}{4} - \frac{1}{9}} \\ = \frac{36}{64} \times \frac{12}{5} \\ = \frac{27}{20}$$

4. (3)

Bohr's formula for angular momentum of an electron in an orbit is given by

$$L = \frac{n\hbar}{2\pi}$$

For the first orbit, $n = 1$ and the angular momentum is $L_1 = \frac{\hbar}{2\pi} = L$

For 2nd orbit

$$L_2 = \frac{2\hbar}{2\pi} = 2L$$

Hence, the change in angular momentum is

$$\Delta L = 2L - L = L$$

5. (3668)

Using Rydberg's formula,

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For the lowest wavelength in Lyman series, $n_1 = 1$, $n_2 = \infty$.

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = RZ^2$$

For Balmer series, $n_1 = 2$, $n_2 = \infty$

$$\frac{1}{\lambda'} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = \frac{RZ^2}{4}$$

So,

$$\frac{1}{\lambda'} = \frac{1}{4\lambda}$$

$$\Rightarrow \lambda' = 4 \times 917 \text{ \AA} = 3668 \text{ \AA}$$

6. (3)

Let the higher state be n .

The number of wavelengths emitted is given by,

$$\frac{n(n-1)}{2} = 6$$

$$\Rightarrow n^2 - n - 12 = 0$$

$$\Rightarrow (n-4)(n+3) = 0$$

$$\Rightarrow n = 4$$

For hydrogen atom,

$$h\nu = 13.6 \left(\frac{1}{1} - \frac{1}{16} \right) = \left(13.6 \times \frac{15}{16} \right) \text{ eV}$$

$$\Rightarrow \nu = \frac{E}{h}$$

$$= \frac{13.6 \times 15 \times 1.6 \times 10^{-19}}{16 \times 6.62 \times 10^{-34}}$$

$$= 3 \times 10^{15} \text{ Hz}$$

7. (3)

Energy of Hydrogen like atoms is, $E = \frac{-13.6Z^2}{n^2} \text{ eV}$.

Since Helium is in the first state, $Z = 2$, $n = 2$.

The energy of He^+ ion is,

$$E = -\frac{13.6 \times 2^2}{2^2} = -13.6 \text{ eV}$$

8. (4)

According to Bohr's postulates, if an electron absorbs a photon with energy equal to the difference between the energies of an excited state and the ground state, it will jump to a higher energy orbital.

Considering that the energy of the colliding electron is 10.2 eV or 12.09 eV, depending on how much energy is lost during the collision. $n = 3$ is the maximum excited state. Hence, there are spectral lines in a total of

$${}^3C_2 = \frac{3!}{2!1!}$$

$$= 3$$

9. (2)

The formula to calculate the radius (r) of any n^{th} Bohr's orbit is given by

$$r = r_0 \frac{n^2}{Z} \dots (1)$$

where, r_0 is Bohr's radius, n is the principal quantum number and Z is the atomic number.

Thus, for Helium ion,

$$r_1 = r_0 \frac{2^2}{2}$$

$$= 2r_0 \dots (2)$$

And, for the Beryllium ion,

$$r_2 = r_0 \frac{4^2}{4}$$

$$= 4r_0 \dots (3)$$

Divide equation (3) by equation (2) to obtain the required ratio.

$$\frac{r_2}{r_1} = \frac{4r_0}{2r_0}$$

$$= 2$$

Hence, $x = 2$.

10. (4125)

The energy of a photon is given by $E = \frac{hc}{\lambda}$.

It is given that $\lambda_1 = 500 \text{ nm}$, $\lambda_2 = 600 \text{ nm}$.

The net energy absorbed is

$$\begin{aligned}\Delta E &= \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = \frac{hc}{10^{-9}} \left(\frac{1}{500} - \frac{1}{600} \right) \\ &= \frac{6.6 \times 10^{-34} \times 3 \times 10^8 \times 100}{500 \times 600 \times 10^{-9}} \\ &= \frac{6.6 \times 3}{30} \times 10^{-19} \text{ J}\end{aligned}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$= \frac{6.6 \times 3}{30 \times 1.6} \text{ eV}$$

$$= 4125 \times 10^{-4} \text{ eV}$$

11. (5)

For λ_1

$$\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \left(\frac{5}{36} \right) \dots (i)$$

For λ_2 ,

$$\frac{1}{\lambda_2} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = R \left(\frac{7}{144} \right) \dots (ii)$$

Dividing (ii) by (i)

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{7}{144}}{\frac{5}{36}} = \frac{7}{20}$$

So,

$$\frac{7}{4n} = \frac{7}{20} \\ \Rightarrow n = 5$$